

LANE A — the confinement engine: the K-type $\leftrightarrow$ stratum projection and its kernel (dedicated look)

The one transition that closes as a *projection with a kernel* — and the kernel is the physics. The map stays Derived; the two physical readings it *induces* are tiered, and the two forward leads are pre-registered, not claimed (C6). Address named before value throughout. Reconnected: the round-4 Šilov theorem ($\lambda_2 > 0 \Rightarrow$ zero boundary value), the $m_1=0$ engine, the $B_2 = \text{SO}(5)$ root system, T2517 (strata addresses).

The map (Derived structure)

- Object: an $\text{SO}(5) \times \text{SO}(2)$ K-type, highest weight (λ_1, λ_2) [entry 5], and the Šilov boundary $\check{S}$ [entry 3]. Verb (v): boundary restriction (the Poisson–Szegő boundary-value map).
- The projection $\Pi_{\check{S}} : A^2(D_{IV^5}) \rightarrow \{\text{functions on } \check{S}\}$. By the round-4 Šilov theorem, a K-type’s boundary value vanishes iff $\lambda_2 > 0$ (non-spherical in the short-root e_2 direction). So:
 - $\ker \Pi_{\check{S}} = \{ \text{K-types with } \lambda_2 > 0 \}$ — the modes that never reach the boundary (confined);
 - $\text{im } \Pi_{\check{S}} = \{ \text{K-types with } \lambda_2 = 0 \}$ — spherical in the short root, reach the boundary (emitted).
- Tier: DERIVED-structure. The projection and the $\lambda_2 > 0$ kernel are the round-4 theorem (banked); this is “closes as a projection with a kernel,” not a full bijection — and the kernel is where the content is.

What it reads (two physical readings, tiered)

1. Confinement (the master mechanism). A mode confined $\Leftrightarrow$ it carries short-root weight ($\lambda_2 > 0$) $\Leftrightarrow$ zero Šilov support. *Reading:* the bulk/boundary split of the domain is the confined/emitted split of the spectrum — a geometric confinement criterion, read by verb (v). Tier: DERIVED as a criterion (round-4); the *identification* with color confinement is a lead (below).
2. $m_1 = 0$ (the lightest mode is massless at leading order). The leading mass operator is supported on the boundary (emitted, $\lambda_2=0$); the confined sector contributes no leading mass. *Reading:* the rank-1 leading mass matrix (only the heaviest generation massive at leading order) is the boundary-supported part. Tier: STRUCTURAL (the link $\lambda_2 \leftrightarrow$ generation-mass is not yet the down-sector Gram computation; do NOT re-run the quark/lepton mass machinery here — mass gate closed).

Two forward leads — PRE-REGISTERED (C6), not claimed

Logged to the reading ledger with expected form + tolerance before any value is read; a null result is reported beside a hit.

lead	pre-registered form	what would confirm	what would refute	tier now
L1 — color confinement is $\lambda_2 > 0$	the color block V_{12} (the real $SO(3)$ triplet, entry 4) carries short-root weight $\lambda_2 > 0$, so color modes are exactly the confined sector	V_{12} 's K-type has $\lambda_2 > 0$ <i>and</i> the $\lambda_2 = 0$ sector is color-singlet	V_{12} sits at $\lambda_2 = 0$ (would reach the boundary — no confinement)	LEAD (open) — a clean geometric statement of color confinement <i>if</i> it lands; else a null
L2 — $m_1 = 0 \Leftrightarrow$ the confined generation	the lightest generation (address $\nu = 0$, Šilov point, T2517) coincides with the $\lambda_2 > 0$ confined kernel at leading order	the $\nu = 0$ stratum mode is in $\ker \Pi_{\check{S}}$	$\nu = 0$ is $\lambda_2 = 0$ (emitted)	LEAD (open) — do NOT compute masses; test only the λ_2 -membership property (the new rule: test the property, not existence)

Guardrail: these are *property tests* (is V_{12} at $\lambda_2 > 0$? is $\nu = 0$ in the kernel?), not mass computations. The mass gate stays closed; no lepton re-run, no Koide. If a lead lands it is a *structural* statement (which modes are confined), not a spectrum claim.

Why this matters for the atlas

The confinement projection is the atlas's first transition whose kernel is load-bearing — the “private information that never reaches the boundary” (the Bergman $\leftrightarrow$ Szegő $\ker \Pi$, made concrete). It is the linear-algebra picture of the master mechanism: bulk = confined = $\lambda_2 > 0$; boundary = emitted = $\lambda_2 = 0$. Every downstream confinement reading is a reading of *this one projection* on D_{IV}^5 .

Lyra, 2026-08-21 (Lane A, confinement engine). The K-type $\leftrightarrow$ Šilov-stratum transition closes as a projection $\Pi_{\check{S}}$ with $\ker \Pi_{\check{S}} = \{\lambda_2 > 0 \text{ K-types}\}$ (round-4 theorem, DERIVED-structure): confined = short-root-carrying = zero Šilov value; emitted = $\lambda_2 = 0$ = spherical, reaches the boundary. Induced readings: confinement criterion (DERIVED), $m_1 = 0$ leading-mass (STRUCTURAL). Two forward leads PRE-REGISTERED, not claimed: L1 color-confinement = V_{12} at $\lambda_2 > 0$; L2 $m_1 = 0 \Leftrightarrow \nu = 0$ in the kernel — both PROPERTY-tests (not mass computations; mass gate closed). Address before value; the kernel is the physics. Nothing pushed; CP existence-only.